

Python Code

```
1  import pandas as pd
2  import numpy as np
3
4
5  data = {
6      'Student_Name': ['Vijay', 'Arun', 'Kiran', 'Priya'],
7      'Mark': [85, 90, 78, 88],
8      'City': ['Chennai', 'Madurai', 'Coimbatore', 'Trichy']
9  }
10
11  df = pd.DataFrame(data)
12
13  print("Original Data")
14  print(df)
15  print("\nColumn Names:", df.columns.tolist())
16
17
18  print("\nAdding Grade Column")
19  df['Grade'] = ['A', 'A+', 'B+', 'A']
20  print(df)
21
22
23  print("\nRemoving City Column")
24  df_drop = df.drop(['City'], axis=1)
25  print(df_drop)
26
27
28  print("\nRenaming Columns")
29  df_rename = df.copy()
30  df_rename.rename(columns={
31      'Student_Name': 'Name',
32      'Mark': 'Score'
33  }, inplace=True)
34  print(df_rename)
35
36
37  print("\nCreating CSV File")
38  df.to_csv('student.csv', index=False)
39
40
41  df_csv = pd.read_csv('student.csv')
42  print(df_csv)
43
44
45  print("\nCreating Excel File")
46  df.to_excel('student.xlsx', index=False)
47
48
49  df_excel = pd.read_excel('student.xlsx')
50  print(df_excel)
```

```
51
52
53 print("\nType Conversion")
54 df['Mark_Float'] = df['Mark'].astype(float)
55 print(df)
56
57 print("\nShape:", df.shape)
58
```

```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:/Users/vijay/OneDrive/Documents/GEN AI/VIJAY-GEN AI-VL-A7.2.py =====
Original Data
  Student_Name  Mark      City
0      Vijay    85      Chennai
1      Arun    90      Madurai
2      Kiran    78    Coimbatore
3      Priya    88      Trichy

Column Names: ['Student_Name', 'Mark', 'City']

Adding Grade Column
  Student_Name  Mark      City  Grade
0      Vijay    85      Chennai  A
1      Arun    90      Madurai  A+
2      Kiran    78    Coimbatore  B+
3      Priya    88      Trichy  A

Removing City Column
  Student_Name  Mark  Grade
0      Vijay    85    A
1      Arun    90    A+
2      Kiran    78    B+
3      Priya    88    A

Renaming Columns
```

Renaming Columns

	Name	Score	City	Grade
0	Vijay	85	Chennai	A
1	Arun	90	Madurai	A+
2	Kiran	78	Coimbatore	B+
3	Priya	88	Trichy	A

Creating CSV File

	Student_Name	Mark	City	Grade
0	Vijay	85	Chennai	A
1	Arun	90	Madurai	A+
2	Kiran	78	Coimbatore	B+
3	Priya	88	Trichy	A

Creating Excel File

	Student_Name	Mark	City	Grade
0	Vijay	85	Chennai	A
1	Arun	90	Madurai	A+
2	Kiran	78	Coimbatore	B+
3	Priya	88	Trichy	A

```
Type Conversion
Student_Name Mark      City Grade Mark_Float
0      Vijay   85      Chennai   A      85.0
1      Arun   90      Madurai   A+     90.0
2      Kiran   78  Coimbatore  B+     78.0
3      Priya   88      Trichy    A      88.0
```

```
Shape: (4, 5)
```

```
>>>
```